

AI Client Retention Copilot

Architecture & Logic Flow Documentation
Client Retention Portfolio

React + Supabase

Google Gemini AI

Edge Functions

SJ Marketing Control Tower
Version 1.0 · July 2026
Audience: Engineers, PMs, Technical Leads

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1. Executive Summary

The **AI Client Retention Copilot** monitors client delivery health across the portfolio, predicts churn risk, and drives recovery actions. It aggregates signals from **ActiveCollab**, **Control Tower**, task comments, and meeting transcripts — then runs a heuristic scoring engine plus **Google Gemini** analysis to produce actionable health snapshots.

Core question: *Is this client at risk, why, on which projects, and what should we do next?*

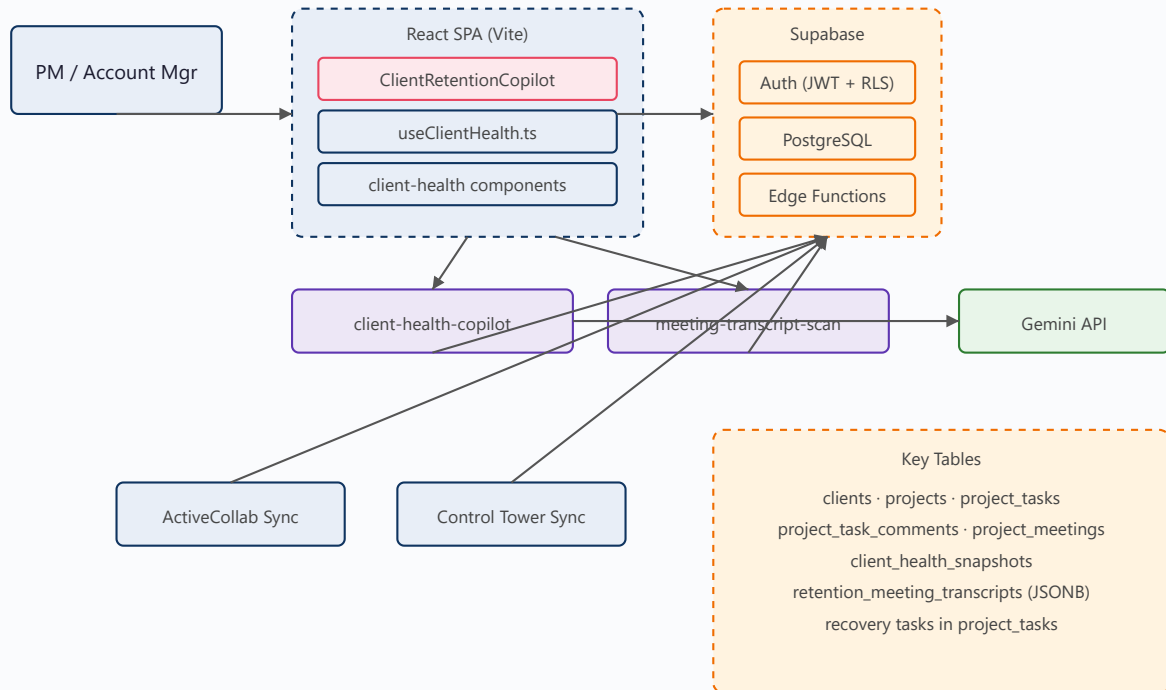
Attribute	Value
Primary route	/client-retention-copilot
Minimum role	pm (Project Manager or above)
Main edge function	client-health-copilot
AI provider	Google Gemini (via Supabase Edge Function)
State management	TanStack Query v5
Persistence	client_health_snapshots + denormalized clients fields

Scope

In Scope	Out of Scope
ActiveCollab & Control Tower tasks, comments, deadlines, meeting transcripts, recovery tasks	HubSpot, Google Analytics, GSC, Slack, Teams, invoices

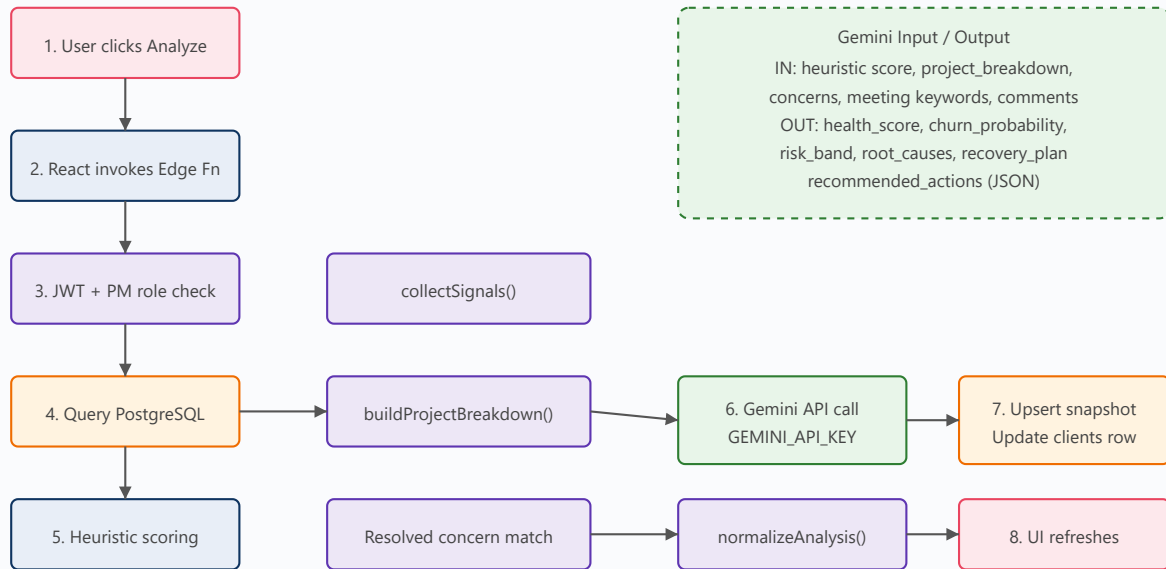
2. System Context Diagram

Figure 1 — End-to-end system context



3. Supabase & AI Integration

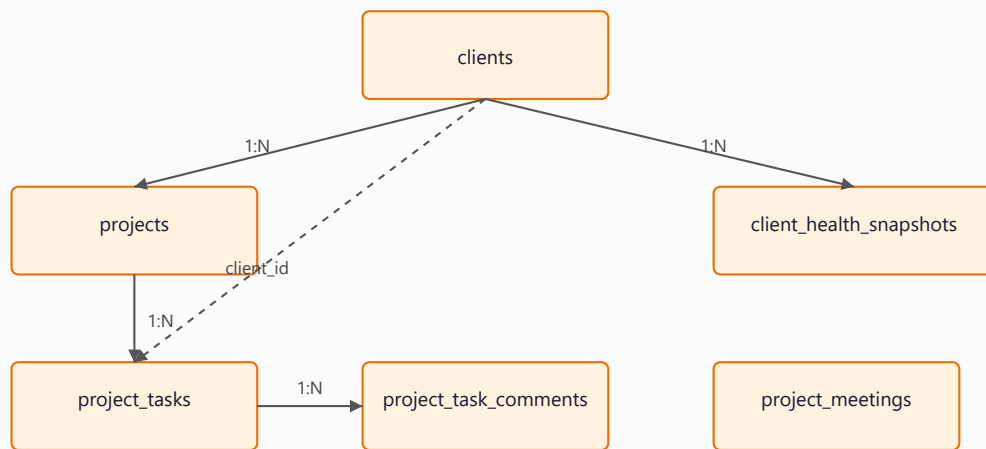
Figure 2 — Supabase + Gemini AI integration flow



Component	Technology	Role
Frontend	React 18 + TanStack Query	Portfolio UI, triggers analysis, displays snapshots
Auth	Supabase Auth (JWT)	Session tokens on every request
API layer	Supabase Edge Functions (Deno)	Signal collection, scoring, AI orchestration
Database	PostgreSQL + RLS	Persistent storage, row-level security
AI	Google Gemini	Structured churn analysis and recovery recommendations
Secret	GEMINI_API_KEY	Stored in Supabase Edge Function secrets

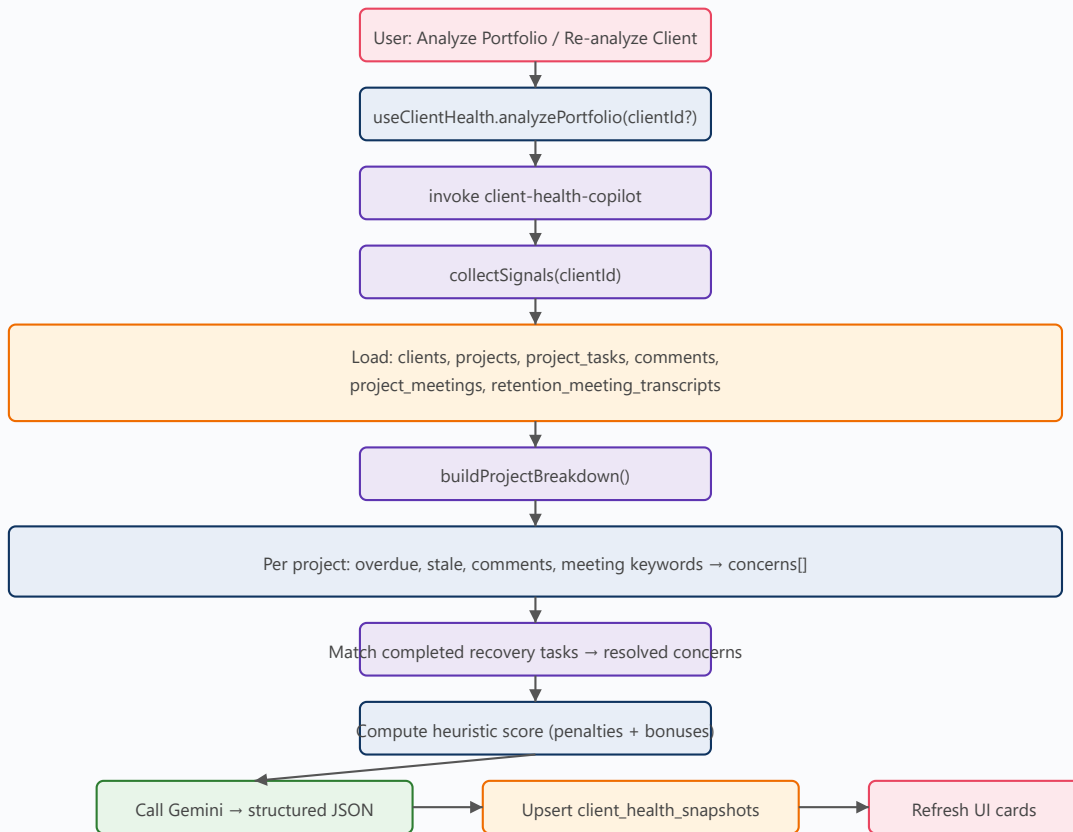
4. Data Model

Figure 3 — Entity relationships



5. Analysis Pipeline (Logic Flow)

Figure 4 — Portfolio analysis sequence



6. Heuristic Scoring Rules

Base score: `clients.satisfaction_score` (default 85). Final score clamped 0–100.

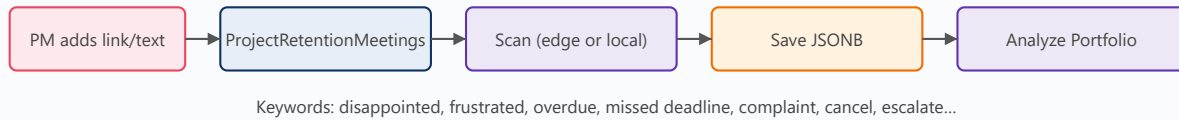
Condition	Adjustment
Each overdue open task	–5 (max –25)
Each stale open task (14+ days no update)	–10 (max –20)
No meeting touch in 21+ days	–15
Negative comment flags	–5 each (max –15)
Meeting negative flags	–5 each (max –15)
Unresolved meeting concern keywords	–4 each (max –20)
3+ tasks due within 7 days	–5
Each resolved concern (completed recovery task)	+6 (max +30)

Risk Bands

Band	Score Range	Churn Window
healthy	85+	90 days
stable	70–84	90 days
watch	50–69	90 days
critical	30–49	45 days
immediate	<30	30 days

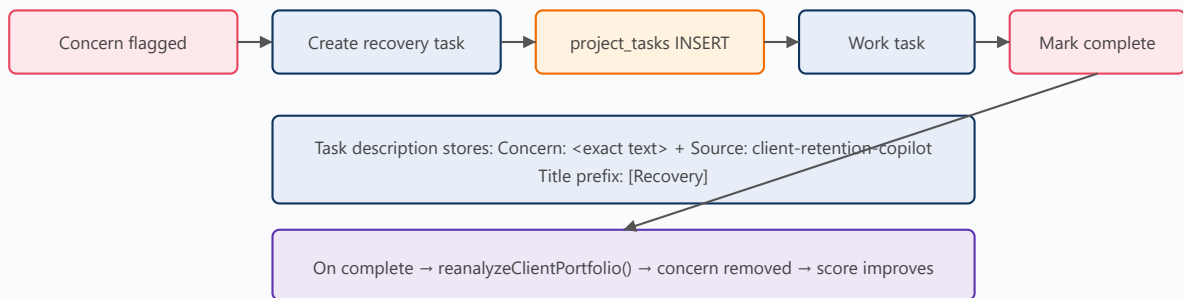
7. Meeting Concern Pipeline

Figure 5 — Meeting transcript → concern keywords → score impact



8. Recovery Task Lifecycle

Figure 6 — Concern → recovery task → score revision



9. Frontend Architecture

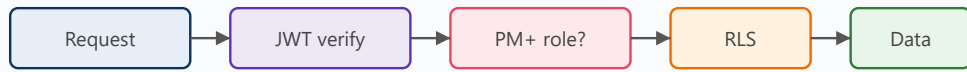
Route	Component	Purpose
<code>/client-retention-copilot</code>	ClientRetentionCopilot.tsx	Portfolio dashboard, analyze all clients
<code>/client-retention-copilot?client=uuid</code>	Same	Deep link to client detail panel
<code>/clients/:slug</code>	ClientDetail.tsx	Retention Portfolio + Run Analysis
<code>/projects/:slug/details</code>	ImportedProjectDetail.tsx	Meetings tab, transcript input

Key React Query Keys

- `client-health-snapshots` — latest analysis per client
- `recovery-tasks` — recovery task list
- `recovery-tasks-summary` — pending/ongoing/completed counts

10. Security & Authorization

Figure 7 — Access control flow



11. Deployment Checklist

1. `supabase db push` (or `--include-all` if migrations out of order)
2. `supabase functions deploy client-health-copilot`
3. `supabase functions deploy meeting-transcript-scan`
4. Set `GEMINI_API_KEY` in Supabase Edge Function secrets
5. Log in as PM+ and run **Analyze Portfolio** on `/client-retention-copilot`

12. Key File Reference

Area	File Path
Portfolio page	<code>src/pages/ClientRetentionCopilot.tsx</code>
Data hook	<code>src/hooks/useClientHealth.ts</code>
Analysis engine	<code>supabase/functions/client-health-copilot/index.ts</code>
Transcript scan	<code>supabase/functions/meeting-transcript-scan/index.ts</code>
Concern scan logic	<code>supabase/functions/_shared/meeting-concern-scan.ts</code>
Meeting UI	<code>src/components/projects/ProjectRetentionMeetings.tsx</code>
DB schema	<code>supabase/migrations/20260227000000_client_health_copilot.sql</code>
Architecture (Markdown)	<code>.agent/System/features/client-retention-copilot-architecture.md</code>